Department pf Computer Science

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B.Sc.(ECS)-II

Advanced Python Programming Practical Assignment-1

1. Write a python program to print odd numbers of a list by using list comprehension.
2. Consider a string “I live in sangola” write a python program to print words less than 5 Characters by using list comprehension.
3. Write a python program which prints numbers between 10 to 50 which are divisible by 3 by using tuple comprehension.
4. Write a python program to demonstrate dictionary Comprehension.
5. Write a python program in which a variable can be accessed everywhere in program.
6. Write a python program to demonstrate nonlocal keyword.
7. Write a python function which accepts another function as an argument as well as returns another function from it’s definition.
8. Write a python function to find given number is prime or not.(Use type hinting)
9. Write a program to print numbers which are divisible by 3 in a range 10 to 50 in reverse manner by using filter function.
10. Write a map function by using lambda function to print square of each element present in a list.
11. Write a python program to demonstrate reduce function.
12. Write a program to create an iterator which prints elements of an iterable.
13. Write python program to demonstrate

    a) count() b) cycle() c) repeat()

14. Write python program to demonstrate

    a) accumulate() b) groupby() c) chain()

15. Consider a list [1,2,3,4,5]
    Make chunks of size 3 of above list by using a function of appropriate Terminating iterator category.
16. Write python program to demonstrate
    a) dropwhile()   b) filterfalse()   c) isslice()
17. Consider a list of temperature of a week in degree celsius [42,46,41,42,44]
    find the temperature difference of each next day with it's previous day by using appropriate Terminating iterator category.
18. Write python program to demonstrate
    a) permutations()   b) combinations()   c) product()
19. Write a python program of Arbitrary arguments which accepts a dictionary of arguments.
20. Write a program to create a new object and stores reference to the original elements.
21. Write a program to create a new object which copies all nested objects within the original elements.
22. Write a python program to match a pattern of a mobile number which starts from 8 or 9.
23. Write a regex pattern by using special sequence to extract only digits from input.
24. Consider a string "sangola mahavidyala sangola Tal:sangola Dist:solapur "
    Replace all white spaces by "*" use appropriate regex method.
25. Write a Python program to find all words starting with 'a' or 's' or 'r' in a given string.